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What shapes the impact of environmental regulation on competitiveness? Evidence from Executive Opinion Surveys

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ABSTRACT

In diametrical opposition to standard predictions, the theory of lead markets conjectures potential competitiveness gains from environmental regulation. Evidence for the actual impact direction and its determinants has however so far been inconclusive. Examining different regulation characteristics and distinguishing between their long-term and short-term effects brings the two seemingly contradictory concepts in line. Based on Executive Opinion Surveys that are annually conducted by the WEF and the IMD, country-level competitiveness effects are evaluated and the determinants are estimated in a panel regression. The estimation results call for a flexible regulation design that allows for different ways of achieving compliance. The fact that regulatory pressure is found to be strongly associated with a more positive perception of long-term impacts lends support to the induced-innovation-hypothesis. While controlling for potential effects from affluence, market size and trade openness, the impact on long-term competitiveness seems also to be affected by environmental quality institutions.

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1. Introduction

In which way a country's competitiveness is affected by environmental regulation continues to be controversially debated. Different schools of economic thought produce diametrically opposite expectations about the direction of regulatory impact. While standard economic models predict

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negative effects, evolutionary approaches of induced innovation defend the potential of positive impacts. This study confirms the standard notion of negative compliance costs effects and provides empirical support for the induced innovation hypothesis by showing that the direction of regulatory impact depends on the considered time horizon.

Under neoclassical assumptions the strengthening of national environmental regulation is supposed to erode international competitiveness, because imposing new constraints on business activity would create additional costs at the firm-level. More efficient investments would be crowded out by expenditures to reduce pollution externalities in compliance with regulation (Jaffe et al., 1995; Palmer et al., 1995), putting domestic firms at a disadvantage relative to their international competitors. According to the Pollution Haven Hypothesis, firms would eventually leave the country for less strict (and hence less expensive) regulatory regimes. Although the effect has so far been difficult to demonstrate empirically, concerns about the loss of competitiveness are immediately raised when governments seek to strengthen environmental regulation.

The opposite proposition, namely that international competitiveness may actually be increased following regulation-induced improvements in product quality and production processes, has been more emphatically articulated over the last two decades by proponents of a neo-Schumpeterian approach. Porter (1991) claims that environmental regulation will enhance a country's competitiveness, based on the assumption that there are fundamental market imperfections that can be overcome by regulatory pressure. The Porter Hypothesis, according to which properly designed environmental regulation can create incentives for innovation that may offset the related costs (Porter and van der Linde, 1995), most prominently challenged the conventional wisdom.

Similarly, the concept of lead markets for environmental innovations (Beise and Rennings, 2003, 2005; Jacob et al., 2005) conjectures potential competitiveness gains at country-level from implementing ambitious national environmental policies. According to this view, the adaptation of a local production system to a stringent regulatory landscape can create early-mover-advantages for the national economy by anticipating global trends and directing technological change towards emerging fields of eco-innovation.

With very few exceptions (Brunnermeier and Cohen, 2003; Lanoie et al., 2008) previous empirical studies usually relied on contemporaneous comparisons. By distinguishing between long-term and short-term competitiveness effects of environmental regulation the two seemingly contradictory concepts are brought in line. Since technological change takes time to become effective, the immediate impact on competitiveness is likely to be different from the long-term effect of environmental regulation. It is reasonable to expect that competitiveness is primarily affected by compliance costs at the beginning, while induced innovation effects dominate the impact later on.

Many empirical studies lack a clear definition of the underlying concept of international competitiveness. In this study, the relevant competing entity is the national economy composed of many firms in different sectors. It is considered to be competitive if it sustains a large number of businesses ready to successfully compete on world markets. Although there is a growing evidence that countries are less competitive in certain sectors with a stringent environmental regulation (Ederington, 2010), at the country-level this could be compensated by competitiveness gains in other sectors. Putting a higher price on pollution could actually foster the transition of the whole economy by promoting innovation through new market entrants, by strengthening less polluting firms and by developing a strong environmental industry (Feess and Mühlheußer, 1999).

The general business response to regulatory pressure may also hinge on certain framework conditions. The realization of competitiveness gains likely depends on the ability of domestic firms to timely draw on economies-of-scale and learning curve effects and to consequently set dominant designs in emerging green markets. Reliable market information on environmental performances may be an important precondition for the ability of businesses to effectively translate cleaner production achievements into market success. Hence, the overall performance of the national innovation system and the prevalence of certain quality institutions may significantly contribute to the explanation of observable country differences.

This study contributes to the long-lasting debate by providing novel insight on how the regulatory impact on competitiveness is affected by the instrument design of environmental policy as well as by relevant framework conditions. Opinion data from two annually conducted expert surveys allows for

a differentiated comparison of relevant regulation design characteristics and the associated competitiveness effects across a relatively large sample of 43 countries over a period of 5 years between 2000 and 2004.

The evaluation of a country's environmental regulation is not restricted to its level of stringency, but also includes assessments of its adequate enforcement and the flexibility of regulatory instruments. Average survey responses of local business experts to differently framed questions about the effect of environmental regulation on the competitiveness of businesses serve as impact measures with different time horizons. Using an innovation capability index and an indicator for the diffusion of environmental certifications, the role of framework conditions in shaping these effects is explored. Indicators of affluence, market size, trade openness and the participation in international environmental agreements serve as control variables in the regression.

The estimation results reveal that a reportedly stringent and well-enforced environmental regulation has a negative effect on the perceived short-term impacts on competitiveness, but in accordance with the induced-innovation hypothesis improves the perceived long-term effects. A regulatory design that is reportedly open for different ways of achieving compliance improves the perceived regulatory impact on competitiveness in general. Furthermore, attention is drawn to the complementary role of quality institutions, since the widespread certification of environmental management systems in a country is associated with a better perception of how environmental regulation affects long-term competitiveness.

The remainder is structured as follows. In the next section an overview of the related empirical literature is given, from which hypotheses about the influence of regulatory design characteristics, innovation capabilities and quality institutions are derived in Section 3. The subsequent sections describe the data set and the applied methods of analysis. Section 6 presents and discusses the results which are summarized in the final conclusion.

2. Related empirical literature

Econometric studies of the link between environmental regulation and competitiveness differ considerably as to the choice of the dependent variable, the measure of environmental regulation, the use of control variables, the unit of analysis and the level of industry aggregation. Reflecting the theoretical controversy, two large streams of empirical literature can be distinguished.

The first stream is primarily concerned with changes in trade patterns and industry location decisions after increases in the stringency of regulation. The impact of environmental regulation has been searched for in the levels and growth rates of productivity (Gray and Shadbegian, 1993; Lanoie et al., 2008; Greenstone et al., 2012), changes in trade performances (Tobey, 1990; van Beers and van den Bergh, 1997; Xu, 2000; van Beers and van den Bergh, 2003; Mulatu et al., 2004; Levinson and Taylor, 2008; Marconi, 2012; Costantini and Mazzanti, 2012), the attractiveness for investments (Xing and Kolstad, 2002; Javorcik and Wei, 2008; Waldkirch and Gopinath, 2008; Quiroga et al., 2009; Millimet and Roy, 2011) and the relocation of (polluting) industries (Lucas et al., 1993; Rauscher, 1995; Gray, 1997; Mani and Wheeler, 1998; Brunnermeier and Levinson, 2004). Most of the studies that examined the effect of environmental regulation on international trade performance did not detect any significant negative effect. The evidence for the actual relocation of industries due to environmental costs is also scant.

Based on the assumption that innovation activity would create capabilities that are needed to sustain growth in an internationally competitive environment (Cantwell, 2005) and that technological capabilities and investments in new production capacity would be the main factors influencing international competitiveness (Fagerberg, 1988), the second stream of empirical literature is mainly concerned with the effect of environmental regulation on innovation. These studies have focussed on R&D expenditures (Jaffe and Palmer, 1997; Johnstone and Labonne, 2006; Arimura et al., 2007), patent counts (Popp, 2002, 2006; Brunnermeier and Cohen, 2003; Johnstone et al., 2010; OECD, 2011; Walz et al., 2011) or innovation survey responses (Frondel et al., 2007; Rennings and Rexhäuser, 2011; Rexhäuser and Rammer, 2011; Lanoie et al., 2011; Horbach et al., 2012) as dependent variables. This kind of studies can be further subdivided into those examining the effect of regulatory pressure and those asking for the influence of regulatory design on the inducement of technological innovation.

Being of the first to analyze econometrically whether the stringency of environmental regulation may spur innovative activity, Jaffe and Palmer (1997) do not find any significant effect. First evidence for such induced innovation has later been provided by Newell et al. (1999). Also Brunnermeier and Cohen (2003), Popp (2006) and Johnstone et al. (2010) find that the introduction of different environmental policies positively impacts on certain patenting activities. According to Ambec et al. (2013), the evidence that stricter regulation leads to more innovation is fairly clear and well-established, while the evidence of a positive impact on overall business performance is mixed. It remains unclear whether regulatory pressure positively affects overall country-level competitiveness.

Porter and van der Linde (1995) stress the importance of a proper regulation design for its innovation effect, but scholarly opinions differ substantially in what are considered appropriate environmental policy characteristics. While some scholars either claim that there is no best policy instrument (Kemp, 1997; Klemmer et al., 1999) or that stringency is much more important than instrument design (Frondel et al., 2007; Arimura et al., 2007), others argue that, given a certain stringency level of environmental regulation, the policy instrument type has an important effect on eco-innovation (Jänicke et al., 2000; Johnstone and Labonne, 2006; Jänicke and Lindemann, 2010; Walz et al., 2011; Lanoie et al., 2011). A recent study by the OECD (2011), which uses cross-country time series data, hints at positive effects of the stringency, predictability and flexibility of environmental regulation on environmental patenting activities. Overall, there is weak support for the hypothesis that flexible market-based instruments are better than prescriptive policies. Environmental regulation certainly stimulates certain innovation activities, but whether the induced technological change actually translates into a competitive advantage at country-level and which factors have a bearing on that remains an open question.

Because of the large amount of empirical studies, the reader is referred to the numerous literature surveys (Dean, 1992; Jaffe et al., 1995; Rauscher, 1997; Jenkins, 1998; Mulatu et al., 2003; Brunnermeier and Levinson, 2004; Copeland and Taylor, 2004; Iraldo et al., 2011; Costantini and Mazzanti, 2012; Ambec et al., 2013) in order to get a more comprehensive overview of the available evidence. Empirical studies of the impact of inter-country differences of environmental regulation on inter-country competitiveness differences are relatively few compared to firm-level or sector-level approaches. According to Iraldo et al. (2011), common indicators used to compare competitiveness between countries are GDP or GNP (Jorgenson and Wilcoxen, 1990), per capita GDP (Esty and Porter, 2001) or international trade flows (Mulatu et al., 2004). Dean (1992) and Jenkins (1998) after reviewing the early evidence conclude that there is no clear relationship between environmental regulation and competitive performance. But more recent literature reviews (Iraldo et al., 2011; Ambec et al., 2013) also come to the conclusion that so far the empirical evidence on the competitiveness effects of environmental regulation is still inconclusive, often anecdotal and usually restricted to highly developed economies.

The lack of evidence concerning the actual effect of environmental regulation must be understood in view of its crucial dependence on hard-to-measure institutional conditions, as well as the possibility that the impact keeps changing over time. Using a meta-analysis Mulatu et al. (2003) demonstrate the pivotal role of certain methodological features as to the level of analysis, the measure of regulatory stringency, the definition of competitiveness and the inclusion of developing countries in the sample for explaining divergent empirical results. Compared to other methodological choices, Brunnermeier and Levinson (2004) highlight the importance of controlling for unobserved heterogeneity and potential endogeneity of the stringency of environmental regulation. They assert that cross-sectional studies tend to reject the pollution haven hypothesis, while more recent studies using panel data to solve the problem of unobserved heterogeneity find some evidence of a moderate pollution haven effect. The importance of considering time horizons has been demonstrated by Lanoie et al. (2008) who show that the impact of environmental regulation on productivity becomes less detrimental and even positive when allowing for dynamic effects using lagged variables.

3. Hypotheses

Based on the theoretical literature, environmental regulation is expected to cause effects on country-level competitiveness. Cross-country analyses of regulatory impacts are however difficult

due to many methodological challenges. The puzzling ambiguity of the existing evidence may be due to (1) a changing impact direction over time and (2) a dependence of the impact on non-regulatory framework conditions that are usually omitted in empirical tests.

In accordance with the induced innovation concept, we expect that in short-term country-level competitiveness is primarily affected by the imposed costs of compliance, while the resulting improvements in products and production processes will only prove beneficial in the longer run. Our hypotheses about the impacts of environmental regulation therefore have to clearly distinguish between short-term competitiveness effects via compliance costs and long-term competitiveness effects via innovation incentives. Furthermore, since the positive long-run impact is supposed to result from induced innovation, we have to consider framework conditions that may have an influence on technological change.

3.1. Regulatory design

As the compliance with more ambitious environmental standards is more demanding and more difficult to achieve, raising the level of regulatory standards tends to increase the costs, the number of concerned actors, the need and the urgency to react. An adequately enforced stringency should thus increase compliance costs, but also increase the pressure to innovate.

Many empirical studies focus exclusively on the the competitiveness effect of the stringency of environmental regulation, although evidently other regulatory characteristics also matter. For instance, there can be a large discrepancy between environmental legislation and its actual enforcement. Consistently and fairly enforced environmental regulations eliminate the possibilities of ignoring or bypassing the law. A proper enforcement makes regulatory stringency effective and the consequences of regulation more predictable. Predictability again helps innovation activity, permitting firms to adopt a stable and long-term oriented R&D strategy.

Regulatory pressure that results from stringency and proper enforcement is thus a double-edged sword. With regard to the short-term and long-term effects, our hypotheses regarding the effects of regulatory pressure on a country's competitiveness are as follows:

H1 *(Short-term effect of regulatory pressure).* For strictly and inevitably imposing compliance costs upon domestic companies, regulatory pressure worsens the competitiveness effect of environmental regulation in the short-term perspective.

H2 *(Long-term effect of regulatory pressure).* For creating strong and stable incentives to improve products and processes for domestic companies, regulatory pressure improves the competitiveness effect of environmental regulation in the long-term perspective.

Another important characteristic of environmental regulation relates to the openness of regulatory design to allow for different ways of complying. According to the theory, the less prescriptive and the more performance-based regulation is designed, the more different options it leaves for cost-effectively complying with regulatory standards and the more options it opens up for the development and application of new technological solutions.

Accordingly, our hypotheses regarding the effects of regulatory openness, which reflects less prescriptive and more performance-based standards, are as follows:

H3 *(Short-term effect of regulatory openness).* For leaving more options to cost-effectively comply with given national standards, regulatory openness improves the competitiveness effect of environmental regulation in the short-term perspective.

H4 *(Long-term effect of regulatory openness).* For leaving more opportunities for feasible and rewarding innovative solutions, regulatory openness improves the competitiveness effect of environmental regulation in the long-term perspective.

3.2. Non-regulatory framework

The responsiveness of an economic system to the tightening of environmental regulation also depends on non-regulatory framework conditions. Already since the end of the 1980s, the importance of so-called social or absorptive capacities for technological development (Abramovitz, 1986; Cohen et al., 1990) has been discussed, and was further emphasized by recent research into economic catching-up processes (Fagerberg and Godinho, 2005; Fagerberg and Shrolec, 2008). What happens when innovation incentives are increased through environmental regulation depends on the prevalence of certain social capabilities for the successful development and diffusion of new technologies in the market as well as the existence of linkages for coordination and knowledge sharing within the network of national innovation system actors (Peuckert, 2011).

If the conditions for domestic innovation are not met, the competitiveness effect of environmental regulation remains limited to the short-term impact of compliance costs. Instead of improving product and processes by inventive activities, regulated firms then may simply buy expensive abatement technologies from abroad. Country differences in social capabilities that are key to sustainable technological change should therefore explain international differences in the regulatory impact on competitiveness in the long run, while being neutral in the short-term perspective.

H5 *(Short-term effect of innovation capabilities).* For not systematically changing compliance costs, innovation capabilities do not have an influence on the competitiveness effect of environmental regulation in the short-term perspective.

H6 *(Long-term effect of innovation capabilities).* For increasing the responsiveness of the economic system to innovation incentives, innovation capabilities improve the competitiveness effect of environmental regulation in the long-term perspective.

While the role of the national system of innovation for technological change is generally recognized, only recently economists have started to analyze the impacts of quality infrastructure on innovation (Guasch et al., 2007; Racine, 2011). It shall be argued that the market success of environmental innovations crucially depends on the existence of institutional arrangements that allow the accurate and reliable assessment and the sharing of quality-related information through signaling and monitoring schemes. The institutional complex which is required to overcome the relevant information asymmetries comprises services in the fields of metrology, standardization, conformity assessment, certification and accreditation and has been coined *quality infrastructure* (Sanetra and Marbán, 2007; Peuckert and Gonçalves, 2011; Gonçalves and Peuckert, 2012).

For a successful formation of green technology markets, environmental performance has to be established as a differentiation tool that helps firms to maintain or increase their market shares. Since improvements in environmental performance are often related to cleaner production processes, usually they cannot be directly observed by the consumers. The actual green quality may be hard to assess and eco-products (for instance, green electricity) frequently suffer from informational problems that are typical of credence goods. To create green competition, a complex system of institutions is required that allows regulators and market participants to accurately sanction environmental performances.

Quality infrastructure services support environmental performance in the following way: metrology provides the necessary instruments and methods for a traceable and reliable measurement of environmental performances; standardization through the definition of relevant reference units establishes the link to the environmental preferences of stakeholders; these units serve as criteria for green quality in the process of conformity assessment; certification is used to signal compliance with environmental standards to market actors and regulators; and accreditation gives credibility to these environmental claims by ensuring the competence of certifiers and testing laboratories.

Shaping the rules, norms and values that govern market transactions by making differences in the respective quality dimension visible, measurable and commensurable, a green quality infrastructure is fundamental to establishing better environmental performance as a relevant market selection criterion. With prevailing quality institutions, market forces may act similarly to environmental regulation as innovation-inducing incentives to reduce the environmental externality.

H7 *(Short-term effect of quality infrastructure).* For increasing the market pressures to comply, quality infrastructure worsens the competitiveness effect of environmental regulation in the short-term perspective.

H8 *(Long-term effect of quality infrastructure).* For helping the formation of markets for green innovation, quality infrastructure improves the competitiveness effect of environmental regulation in the long-term perspective.

4. Data

A thorough econometric analysis of the impacts of environmental regulation on competitiveness has to overcome fundamental difficulties. First, the overall impact assessment crucially hinges on the relevant time horizon of the evaluator. Theoretically, even an impact reversion from short-term to long-term perspective is possible, since compliance costs are expected to initially reduce domestic business performances, whereas any potential gains in country-level competitiveness from induced innovation take time before becoming effective. Secondly, to measure potential determining factors, like the various dimensions of environmental policy characteristics, on a globally comparable scale is not trivial at all. Thirdly, relevant framework conditions, like social capabilities for environmental innovation, differ substantially across the world. Without controlling for these effects, it is questionable whether any results can easily be generalized.

4.1. Data set

The empirical approach copes with these challenges by using a panel set of subjective data for a relatively large sample of countries at different levels of income. To evaluate the characteristics and the respective effects of environmental regulation across countries, data are used from two Executive Opinion Surveys, which are regularly conducted by the World Economic Forum and by the Institute of Management Development. Both are carried out annually on a global scale to capture the opinions of business leaders on issues for which hard data is scarce or non-existent. Combining the two data sets yields a fully balanced sample of 43 countries[1] over 5 years between 2000 and 2004.[2]

The data used in the analysis origine from Likert-type survey questions. WEF and IMD annually report the average responses for each country. The sample sizes on which these country values are based vary according to the size of the economy. The WEF survey relies on about 10,000 respondents per year, which corresponds to an average of about 90 per country. The IMD also receives more than 4000 responses with an average of about 70 per observation unit. In order to ensure statistical representativeness, efforts are made that the sample of respondents reflects a cross-section of the respective national business sectors regarding the share of production by industry, company size and the range of different company types (domestic, foreign or partly state-owned). More information about the composition of respondents in each year are given in the respective reports of the WEF and the IMD.

The surveys include explicit questions about the effects of environmental regulation on competitiveness. The per country average responses in each year are used as measures of the regulatory impact. It is important to note that the surveys ask for the *effects* of regulation, not for *competitiveness* as such. Using these subjective assessments mitigates the problems of endogeneity or reverse causation, because in doing so the causal attribution of the competitiveness effect to environmental regulation rests on the expertise of local business leaders not merely on statistical correlations. As a drawback, the magnitude of changes in competitiveness cannot be quantified in absolut terms. The empirical information remains limited to qualitative inferences on the direction of influence.

[1] Austria, Australia, Belgium, Brazil, Canada, Chile, China, Colombia, Czech Republic, Denmark, Finland, France, Germany, Greece, Hongkong, Hungary, Iceland, India, Indonesia, Ireland, Israel, Italy, Japan, Korea, Malaysia, Mexico, Netherlands, New Zealand, Norway, Philippines, Poland, Portugal, Russian Federation, Singapore, South Africa, Spain, Switzerland, Sweden, Turkey, Thailand, USA, United Kingdom, Venezuela.

[2] According to World Bank data, the 43 selected countries together represented more than two thirds of the world's total population and accounted for more than 90% of the world's gross domestic product over the period of 2000–2004.

Table 1
Measures of regulatory impact.

Variable	Survey question
SHRT	"Environmental laws and compliance do not hinder the competitiveness of businesses." (Index from 0 to 10)
LONG	"Complying with environmental standards in your country. . . 1 = significantly reduces competitiveness, 7 = helps long-term competitiveness by encouraging improvements in products and processes."

Table 2
Measures of environmental regulation.

Variable	Survey question
STRN	"How stringent is your country's environmental regulation? 1 = lax compared to most other countries, 7 = among the world's most stringent."
NFRC	"Environmental regulations in your country are . . . 1 = confusing and frequently changing, 7 = transparent and stable."/". . . 1 = not enforced or enforced erratically, 7 = enforced consistently and fairly."/". . . 1 = confusing and enforced erratically, 7 = stable and enforced consistently and fairly.
FLEX	"Environmental regulations in your country. . . 1 = offer no options for achieving compliance, 7 = are flexible and offer many options for achieving compliance."/"Environmental regulations in your country are achieved through. . . 1 = adversarial and legal means, 7 = government-business cooperation and voluntary corporate action."

Having two distinct ratings of the regulatory impact provides the opportunity to analyze their disalignment in detail. Although very similar at first glance, there are subtle differences between both survey questions (see Table 1). The evocative wording of one survey question ("compliance" and "hinder") makes it capture the immediate costs of complying with environmental regulation,[3] while the other question by refering explicitly to the "long-term" consequences from "improvements in products and processes" rather captures the more indirect effects of induced innovation. Although response variables are highly correlated (with a significant positive correlation of .5990), there is still substantial divergence in the answers to be explained. Given the major difference is related to the implied time horizon, analyzing their disparity reveals novel information on the dynamical patterns of the regulatory impact. Interpreting the responses to the one survey question as a measure of the short-term effects (*SHRT*) and to the other as a measure of the competitiveness effects in a long-term perspective (*LONG*), the before mentioned hypotheses can be tested econometrically.

The WEF Report also provides valuable information on the design of environmental regulation. The survey includes questions on the relative stringency (*STRN*), the transparency, fairness, consistency and stability of regulation enforcement (*NFRC*), as well as the openness of environmental regulation for achieving compliance in a flexible and non-adversarial way (*FLEX*).[4] Table 2 gives an overview of these measures, which are used as exogenous variables to explain the impact of environmental regulation on competitiveness.

Other determining factors of competitiveness effects of environmental regulation are supposed to be found in national framework conditions, like a country's general innovation capability and its environmental quality infrastructure, that form relevant social capabilities for the successful creation and diffusion of environmental innovation. These performances are captured by complex indicators described in Table 3.

The *Innovation Subindex* of the WEF Report (*ICAP*) is used as an indicator of general innovation capability. It is composed of both hard and survey data, reflecting many different aspects of the national

[3] As one reviewer has correctly pointed out, the question whether "environmental regulation and compliance do not hinder the competitiveness of business" may lead respondents assessing a lax regulation for their country to automatically assume a low influence on competitiveness.

[4] Some data treatment was necessary to obtain a consistent time series, because several survey questions had been removed, merged, or rephrased between panel waves. The questions on the stability and on the consistency of environmental regulation underwent slight changes and were eventually succeeded by a combined question in 2004. After having verified their high interrelatedness in capturing the proper enforcement of environmental regulation, both variables were averaged for the years before 2004. The same procedure was applied to the variables capturing the flexibility of environmental regulation when both where available for the same year.

Table 3
Measures of relevant framework conditions.

Variable	Description
ICAP	Innovation Subindex, composed of 1/4 innovation survey data (Technological readiness, Firm-level absorption, Company spending on research and development, University/industry collaboration) and 3/4 innovation hard data (Utility patents, Tertiary enrollment)
QUAL	Environmental Quality Infrastructure Index, certification intensity measured as log of the number of ISO 14001 certifications per GDP, rescaled relative to the highest (=1) and the lowest (=0) observed value of the same year

Table 4
Other variables.

Variable	Description
GDP	Gross domestic product, in 10^{12} current US$, Source: World DataBank
POP	Total population, in 10^{8} inhabitants, Source: World DataBank
INC	GDP per capita in 10^{4} current US$, Source: World DataBank
TRD	Sum of exports and imports of goods and services measured as a share of gross domestic product, Source: World DataBank
ISO	Total number of ISO 14001 certifications per country, in 10^{3} certifications, Source: ISO Survey 2011
IEA	Participation in 11 selected international environmental agreements, Source: UNEP GEO Data Portal

innovation system, including technological, financial, educational and relational issues. More than just scientific or technical achievements which are usually measured by patent data and R&D expenditures, this indicator also envisages the economic application of new technologies and the fundamental conditions that create a fertile environment for innovation. The index thus aims to reflect a country's potential to produce a stream of commercially relevant innovations as an important driver of its international competitiveness.

Certifications according to ISO 14001 Environmental Management Systems serve to construct an index of a country's environmental quality infrastructure (*QUAL*). Given the strong systemic interdependencies of quality infrastructure elements, the comparatively extensive use of these certifications is supposed to reflect the general performance of environmental quality institutions. The total number of certifications (*ISO*), as reported by the ISO Survey 2011, is divided by the gross domestic product (*GDP*) to obtain a measure of certification intensity. A log transformation is applied to correct for the high skewness of the value distribution. This certification intensity is put into relation to common certification practice. The index takes values between 0 and 1, depending on a country's performance relative to the minimum and the maximum certification intensity observed among the 43 countries in the respective year.

To complete the data set (Table 4), additional statistical information from the World Bank are used as variables to control for the influences of a country's affluence (*INC*), market size (*POP*) and trade orientation (*TRD*). UN data on the participation in selected international environmental agreements (*IEA*) serve as proxies for coercive pressures from the international community.

4.2. Critical issues of survey data

A common objection against the use of subjective data is the extent to which it may suffer from biases.[5] On the other hand, methodological studies conducted by Nicoletti and Pryor (2006), who

[5] Pryor (2002) describes potential sources of bias: "First, the samples are small and their degree of randomness can be questioned. Second, the respondents were limited to high level business executives, who presumably have experience in dealing with governmental regulations. Given their attitudes toward governmental regulations, however, they may exaggerate or downplay the extent of these regulations. Third, the respondents were asked to judge only their home country so that it is unclear whether they used the same standards of comparison. Given the fact that they are associated in some way with organizations (the WEF or IMD) containing members from many different countries, however, they may have assimilated some common standards of evaluations so that their judgements for their native country are formed with certain comparative experience with other nations."

Table 5
Correlations between variables of environmental regulation.

Variables	*STRN*	*NFRC*	*FLEX*
STRN	1.000		
NFRC	0.925 (0.000)	1.000	
FLEX	0.711 (0.000)	0.832 (0.000)	1.000

compare three different approaches to measure regulation, and by Koch et al. (2010), who use both expert views and objective information about governmental regulations in a cross-country comparison, rather demonstrate that objective and subjective data are highly correlated and apparently capture the same underlying reality with regard to the regulatory framework.

In addition, subjective assessments have important advantages for the comparison of phenomena across very different contexts. For very heterogeneous country samples, hard data is often far from being free of biases. Nominally the same environmental standard can have very different real implications depending on the local circumstances, not to speak about varying enforcement practices. Survey data can put this into perspective. Business leaders are asked to evaluate national conditions relative to the rest of the world. The executives' wealth of international experience allows them to correct for context-specific ideosyncrasies when assessing their working environment. The used data therefore offers a rich and multisided perspective on the different aspects of regulatory designs and impacts for which reliable hard measures are not available.

There is a longlasting debate about how Likert-type survey data are to be analyzed. Statistical methods like factor analysis and hierarchical linear models are based on the assumption of normally distributed, interval-level data. Recognizing that Likert-type items are ordinal in character, some scholars claim that parametric statistics could not be used (Jamieson, 2004). Carifio and Perla (2008) however argue that those who defend this logical position ignore the many studies of robustness that have been done. Also Pell (2005) confirms that parametric techniques can be applied to non-parametric data generated from Likert scales, provided the assumptions are clearly stated and the data is of the appropriate size and shape that resembles a normal distribution.

Despite of being aggregated from Likert-type items, the measures of average survey responses that are used in the following analysis show properties that resemble an interval scale. In fact, based on normality tests, we cannot reject the hypotheses that the variables *SHRT* and *LONG* are normally distributed.[6] It is thus perfectly appropriate to use parametric statistics with all the benefits and power of these higher levels of analysis.

5. Methods

The strong correlation of the environmental regulation measures *STRN*, *NFRC* and *FLEX* (in Table 5) could lead to problems of multicolinearity if all together included in the same econometric estimation. To reduce the dimensionality of these data to only two orthogonal factors without much loss of the original information, principal component analysis is performed.

Applying a standard varimax rotation with Kaiser normalization yields one factor on which the variables stringency (*STRN*) and enforcement (*NFRC*) load high and another with a high factor loading of the flexibility variable (*FLEX*). This makes their interpretation straight-forward: the first factor is labelled Regulatory Pressure (*PRSS*), because it stands for strict regulation requirements based on ambitious and well-enforced environmental standards, the second factor is labelled Regulatory Openness (*OPEN*), because it reflects to which extent the regulatory design allows for different options to

[6] The normal shape of the dependent variable distributions are confirmed by normality tests of skewness and kurtosis as well as the Shapiro–Wilk *W* test for normality.

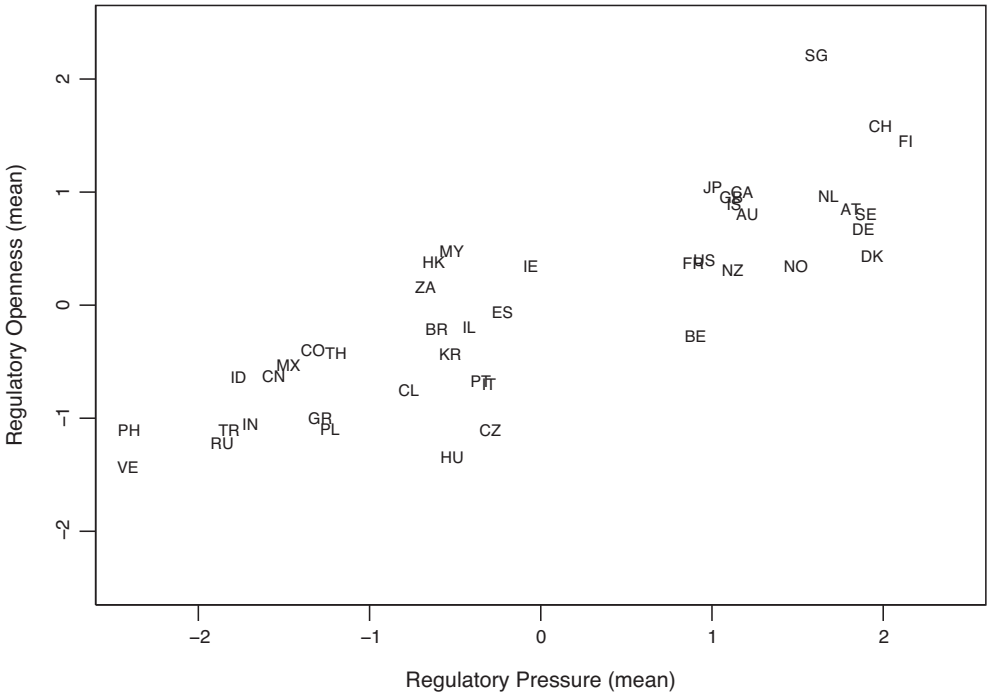

Fig. 1. Environmental regulation factors.

comply with the environmental standards. Fig. 1 depicts the mean factor scores per country over the whole observation period of five years in a scatterplot. The new regulation measures still describe 98% of the original variance.

To analyze how the short-term and the long-term impact of environmental regulation on competitiveness is shaped by the characteristics of regulation, *PRSS* and *OPEN* are used as explanatory variables in two panel regressions with *SHRT* and *LONG* as the dependent variable, respectively. To check for the short-term and long-term influences of national innovation capabilities and the prevalence of environmental quality institutions, *ICAP* and *QUAL* are also included in the set of explanatory variables.

Furthermore, *INC*, *POP*, *TRD*, *IEA* and a set of year dummies are used as control variables in the regression to protect the estimation results against spurious effects from different levels of affluence, unequal market sizes, different trade policies, coercive pressures from the international community or commonly shared time effects from exceptional events. The descriptive statistics for all these variables and the data used to derive them are reported in Table 6.

To control for potential influences from unobserved time-invariant heterogeneity among countries in a panel setting, fixed effects (FE) models are recommended. But, in the absence of such influences, random effects (RE) estimators are more efficient. Therefore both model specifications are estimated. Hausman specification tests indicate whether the estimated coefficients are significantly different, in which case the FE results would be considered. However, if the differences are not significant, there is no reason to believe that RE models are misspecified and the random effects estimates should be used.

Additionally, quantile regressions are performed as robustness checks of the estimation results. This kind of regression is more robust against the influence of outliers than ordinary least squares regressions. The results for the .25, .50 and .75 quantile are reported in Table A.8 in the Annex.

Table 6
Descriptive statistics.

Variable	Obs	Mean	SD	Min	Max
SHRT	215	6.18	0.85	3.92	8.43
LONG	215	4.64	0.60	3.40	6.10
PRSS	215	−0.00	1.39	−2.82	2.49
OPEN	215	0.00	1.01	−2.29	2.60
STRN	215	4.90	1.17	2.50	6.80
NFRC	215	4.53	0.89	2.60	6.55
FLEX	215	4.29	0.55	3.00	5.70
ICAP	215	3.72	1.17	1.89	6.62
QUAL	215	0.63	0.20	0	1
ISO	215	1.17	2.19	0.00	19.58
GDP	215	0.76	1.73	0.01	11.80
INC	215	1.78	1.36	0.05	5.66
POP	215	1.01	2.47	0.00	12.96
TRD	215	0.88	0.69	0.20	4.13
IEA	215	8.89	1.39	5	11

6. Results

The estimation results of the FE and RE regression model specifications with either *SHRT* or *LONG* as the dependent variable are reported in Table 7. Hausman statistics at the bottom of the table confirm that in both cases the estimated coefficients of the RE model are not significantly different from the FE model estimates. Hence, unless otherwise noted, the following discussion of the results will refer to the RE estimations. The relatively high coefficients of determination indicate that the regression models fit the data quite well. Overall, the RE models are able to explain about 40% of the variance in the *SHRT* variable and almost 80% of the variance in the *LONG* variable.

The estimations support the hypotheses H1 and H2 about the influence of regulatory pressure. The expected opposite effects in the short-term and long-term perspective become evident from the inversion of signs of the estimated coefficients for *PRSS* as explanatory variable of *SHRT* and *LONG*. The slightly significant negative coefficient with regard to *SHRT* as opposed to the highly significant positive coefficient with regard to *LONG* imply that in countries where the regulatory pressure is generally assessed to be higher, experts tend to answer less positively to the question whether complying with environmental standards hindered the competitiveness of businesses (H1), but also react definitely more positive when asked for the long-term competitiveness effects of environmental regulation (H2).

Hypotheses H3 and H4 on the positive influence of regulatory openness are clearly corroborated by the estimation results. The regression coefficient for *OPEN* is positive and highly significant in all specifications. This means that in countries where the regulatory openness is generally perceived to be higher, experts tend to answer more positively to both questions regarding the short-term and long-term effects of environmental regulation on competitiveness.

Interestingly, the regression results do not show any significant influence of innovation capability. Although this is in accordance with hypothesis H5, hypothesis H6 on the long-term effects of the responsiveness to innovation incentives is not confirmed. The estimated coefficient for *ICAP* as explanatory variable of the *LONG* variable has the expected positive sign, but it is not significantly different from zero. Hence, innovation capability does not change the perceived impacts of environmental regulation on competitivenss, neither in short-term nor in long-term perspective.

Hypotheses H7 and H8 about the influence of the quality infrastructure are partly confirmed. Against the negative short-term expectations of hypothesis H7, the estimated coefficient for *QUAL* on the *SHRT* variable is not significant.[7] Hence, the assessments of the short-term competitiveness

[7] In contradiction to hypothesis H7, *QUAL* even shows a significantly positive coefficient in the .25-quantile regression reported in Table A.8.

Table 7
Regression results.

[a]	*SHRT*		*LONG*	
	FE	RE	FE	RE
PRSS	−.201*	−.172**	.353***	.290***
	(.1098)	(.0862)	(.0647)	(.0408)
OPEN	.245***	.278***	.084**	.130***
	(.0700)	(.0649)	(.0412)	(.0340)
QUAL	−.040	.308	.094	.335**
	(.5710)	(.3914)	(.3363)	(.1532)
ICAP	.114	.098	.022	.035
	(.0988)	(.0771)	(.0582)	(.0363)
INC	.229	.229***	−.122	−.013
	(.1499)	(.0813)	(.0883)	(.0348)
POP	−.101	.036	−.123	.037***
	(.7290)	(.0356)	(.4294)	(.0116)
TRD	.499	.344**	.094	.083*
	(.4152)	(.1363)	(.2445)	(.0441)
IEA	−.014	−.033	−.032	−.041*
	(.0700)	(.0529)	(.0412)	(.0221)
year				
y_{2001}	−.089	−.085	−.131**	−.105*
	(.0991)	(.0944)	(.0584)	(.0545)
y_{2002}	−.138	−.133	−.047	−.024
	(.1191)	(.1063)	(.0702)	(.0588)
y_{2003}	−.382***	−.390***	−.114	−.127**
	(.1267)	(.1116)	(.0746)	(.0611)
y_{2004}	−.431**	−.421***	.000	−.038
	(.1608)	(.1283)	(.0947)	(.0670)
cons	5.363***	5.367***	4.851***	4.638***
	(1.2701)	(.6261)	(.7481)	(.2670)
R^2				
Within	0.1964	0.1970	0.2694	0.2503
Between	0.3092	0.4728	0.6418	0.9079
Overall	0.2802	0.4141	0.5805	0.7998
Hausman test				
$\Pi > \chi^2$	0.9895		0.4845	

[a] Level of significance: *10%, **5%, ***1%.

effect of environmental regulation are not affected by the widespread use of environmental certification. The significantly positive coefficient of *QUAL* on *LONG* however corroborates hypothesis H8. That is, in countries where environmental quality institutions are widely used, experts tend to assess the long-term effect of environmental regulation more positively.

The control variables show no significant effects in the FE model estimations, but contribute substantially to the explanation of the between country variance in the RE models. The level of income improves the perceived short-term effects of environmental regulation, which could be related to a generally higher demand for environmental quality in more affluent societies. Total population is positively associated with improved long-term assessments, which could be due to a bigger market for the new environmental technology developed in response to regulation. A stronger trade orientation improves both short-term and long-term prospects from environmental regulation, maybe because the compliance with stricter regulations also facilitates the integration in global production chains. Finally, the participation in international environmental agreements is related to more pessimistic evaluations of long-term competitiveness, although this relation is only weakly significant.

6.1. Discussion

The results seem to confirm the standard predictions of negative competitiveness effects from regulatory pressures in the short-term perspective. The sensitivity of these results to the wording is however strikingly demonstrated by the contrary results obtained from the question about long-term competitiveness effects. By interpreting the diametrically opposite findings as resulting from different time horizons, standard economic theory and evolutionary approaches can be brought in line. More evidence is however needed to prove such an inversion of the impact over time.

It was assumed that governments can freely choose the design of environmental regulation. A high positive correlation of Regulatory Pressure (*PRSS*) and Regulatory Openness (*OPEN*) however hints at the existence of common influencing factors. The scope of governments to adequately design regulatory instruments may actually be endogeneously determined by national framework conditions, as for instance the prevalence of innovation capabilities and the existence of an effective quality infrastructure.

In fact, quality infrastructure and environmental regulation are closely coupled through mutual referencing. For instance, certification according to ISO 14001 standards demands adherence to the relevant environmental regulations, thus also acting as an enforcement instrument. Regulations frequently refer to commonly accepted standards or require a conformity assessment to prove compliance. A lacking quality infrastructure therefore may seriously condition the design options available to the regulator. When there are no adequate quality assessment mechanisms available that allow the effective use of performance-based instruments, environmental regulation remains limited to technology-based standards and certain command-and-control instruments.

The apparent insignificance of innovation capabilities for the competitiveness effect of environmental regulation does not preclude an indirect influence via the regulatory design for political economy reasons. The stringency of environmental regulation as well as its flexibility could very well depend on the local availability of certain technological capabilities and implementation options. The flexibility of environmental regulation as well as its enforcement could also depend on the prevalence of certain quality institutions. The structural relationship of regulatory and non-regulatory framework conditions definitely deserves further investigation. More complex structural equations models that are able to describe these kinds of interdependencies should therefore be developed.

7. Conclusion

The difficulty of measuring and comparing regulation characteristics and social capabilities make the quantitative analysis of the interplay between environmental regulation and competitiveness a challenging endeavor. Using survey data from two different sources, the direct influences of regulatory instrument design and non-regulatory framework conditions on the competitiveness effects of environmental regulation over different time horizons have been assessed. The finding of different short-term and long-term effects may explain the inconclusiveness of previous empirical evidence. As expected, the study finds clear evidence that regulatory design characteristics have a strong influence on the impact of environmental regulation on competitiveness. The effect of regulatory pressure is ambiguous and depends on the time horizon of the evaluator. Myopic short-term oriented governments would be recommended to avoid regulatory pressure because of the negative effects compliance costs have on short-term competitiveness. But, in accordance with the lead market concept, there are positive effects to be gained from more stringent and better enforced regulation in the long-term perspective. This is probably due to improvement in products and processes as response to regulatory pressure as suggested by the induced-innovation hypothesis.

The methodological approach that was here chosen does not allow for a direct comparison of both effects. In the end, it depends on the societal weighing of future returns and remains to be investigated from case to case whether the expected positive long-term effects can indeed justify the necessary short-term investments. The important empirical contribution of this study is to clearly demonstrate that there is such a trade-off between short-term and long-term impacts on competitiveness. The study thus points at the importance of clearly defining the relevant time horizon and the relative weighing of future returns when assessing the regulatory impact.

In contrast to regulatory pressure, a flexible design of environmental regulation that is open to different ways of achieving compliance appears to improve its effects on competitiveness, independently of the time horizon. This is consistent with many recent studies that have emphasized the importance of an adequate design of environmental regulation for positive competitiveness effects. The same environmental standard may be achieved through very different technological solutions. Implementing policy instruments that enforce a certain environmental standard without being prescriptive as to how it has to be achieved, help to reduce the compliance costs by allowing for the most cost-efficient solution and at the same time create a space for competition among innovative environmental technologies.

Quality infrastructure has also been found to affect the long-term impact of environmental regulation on competitiveness, pointing to its complementary role for incentivizing environmental innovation. Apparently, a well-functioning environmental quality infrastructure supports directed technological change, as the existence of institutions that enable the measurement, the credible signaling, and the effective monitoring of environmental quality help companies to take advantage of improvements that have been induced by environmental regulation. More effort has yet to go into the empirical assessment of national framework conditions for sustainable development.

This is particularly true for the role that innovation capabilities and quality infrastructure institutions have in enabling governments to design environmental regulation in such a way that its competitiveness effects are optimized. An econometric analysis of these relationships requires the development of more complex structural models. It could however yield better insights into which social capabilities are needed to simultaneously pursue the policy goals of environmental protection and economic development by regulation.

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Appendix A.

See Table A.8.

Table A.8
Quantile regression results.

[a]	*SHRT*			*LONG*		
	q25	q50	q75	q25	q50	q75
PRSS	−.129	−.136	.011	.260***	.268***	.265***
	(.1617)	(.1292)	(.0937)	(.0361)	(.0596)	(.0509)
OPEN	.426***	.461***	.309***	.182***	.158***	.177***
	(.1206)	(.1238)	(.0974)	(.0463)	(.0676)	(.0525)
QUAL	.727**	.302	.333	.494**	.200	.160
	(.4567)	(.5917)	(.4234)	(.2456)	(.2001)	(.1621)
ICAP	−.135	.156	.108	.058*	.069	.028
	(.0983)	(.1726)	(.1045)	(.0343)	(.0705)	(.0514)
INC	.272***	.044	.041	−.037	−.037	.005
	(.0977)	(.1227)	(.1094)	(.0489)	(.0417)	(.0328)
POP	.037	.028	.031	.031***	.033**	.042**
	(.0268)	(.0223)	(.0167)	(.0103)	(.0129)	(.0161)
TRD	.242**	.280**	.241***	.072*	.124*	.057
	(.1065)	(.0864)	(.0790)	(.0418)	(.0688)	(.0721)
IEA	−.138**	−.033	.061	−.040	−.019	−.017
	(.0689)	(.0601)	(.0585)	(.0253)	(.0312)	(.0343)

Table A.8 (*Continued*)

[a]	*SHRT*			*LONG*		
	q25	q50	q75	q25	q50	q75
year						
y_{2001}	−.209	−.080	−.067	−.066	−.148*	−.134
	(.2598)	(.2053)	(.1250)	(.0956)	(.0867)	(.0869)
y_{2002}	−.311***	−.057	−.050	−.063	−.017	−.095
	(.2488)	(.2391)	(.1822)	(.1057)	(.1369)	(.0933)
y_{2003}	−.743***	−.164	−.351**	−.147	−.111	−.182*
	(.2616)	(.2708)	(.1371)	(.1153)	(.1286)	(.0991)
y_{2004}	−.508*	−.338	−.246*	−.018	−.058	−.114
	(.2756)	(.2539)	(.1353)	(.0921)	(.1268)	(.1042)
cons	6.564***	5.533***	6.391***	4.332***	4.424***	4.806***
	(.9559)	(.7833)	(.6335)	(.2234)	(.4910)	(.4586)
R^2	0.2156	0.2496	0.3075	0.5593	0.5743	0.6053

[a] Level of significance: *10%, **5%, ***1%.

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